Suraj Tripathi

Software Engineer 2
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GitHub | **Linkedin** | **Portfolio**

Education

JSS Academy of Technical Education
Computer Science and Engineering B.Tech

PP Savani University
Data Science M.Sc

Experience

Amazon Web Services | **Software Engineer 2** Seattle | September 2024 - Present

- Implemented Option 43 (vendor-specific info) support for Cisco access point devices in amazon DHCP systems, enabling seamless provisioning of 9,691 Cisco devices at scale including TLV sub-option configuration via UI.
- Designed and delivered the DHCP configuration service Integration Canary, a fully automated end-to-end validation system that runs config mutation tests every 10 minutes via AWS Step Functions reducing service issue detection time from days (waiting for customer reports) to under 10 minutes, with automatic real-time failover and deployment rollback, triggered on SEV-2 alarm, eliminating customer-impacting silent failures in DHCP configuration changes.

Amazon Web Services | **Software Engineer 2** Bengaluru, India | Feb 2020 - September 2024

- Designed and developed the full-stack UI for a RAG (Retrieval Augmented Generation) based AI chatbot application for Takenaka Corp (a Japanese enterprise customer), architecting a React + Material UI frontend with AWS Amplify/Cognito authentication (Azure AD SSO), real-time WebSocket chat via Node.js, DynamoDB-backed chat history, and data source filtering delivering a secure, production-grade conversational AI interface enabling employees to query legal and standard documents through natural language.
- Built 2 CI/CD pipelines (front-end and backend) for the Payroll chatbot, reducing release cycle time to under 2 hours and meeting a 24-hour Disaster Recovery SLA contributing to 13% chatbot adoption among Slovakian market employees within 2 months of launch, supported by 6 CloudFormation templates and 2 Python scripts developed to meet evolving customer requirements.

Digital Alpha Platforms | **Software Engineer** Noida | March 2019 - Feb 2020

Built an NLP-powered news summarization system using Python, applying two extractive summarization approaches word frequency scoring and Named Entity Recognition (NER)-based sentence scoring with support for both local article ingestion and live web scraping, demonstrating early applied ML research and hands-on NLP engineering using spaCy, NLTK, and TF-IDF techniques.

Vidooly Media Tech | **Member of Technical Staff** Noida | Jan 2016 - Feb 2019

Worked on NLP and machine learning on various projects which involve keyword extraction, comment classification, and word sense disambiguation to help provide better video recommendation.

Skills

Programming Languages:	C++, Python, Java, Node.Js, Ruby, Go, Typescript, CDK, Javascript, C
Libraries/Frameworks:	React, FastAPI, Django, Flask, CI/CD, Devops, Machine Learning, Deep Learning, LLMs, Spring, JDBC, Hibernate, Vue.Js, Android Studio, React Native, Agile, Claude Sonnet, RAG, SSL/TLS, SSH, Angular, Gradle, Maven, Tomcat, Spring Boot, ODBC, Natural Language Processing, Computer Vision, OCR, Keras, Pandas, Numpy, Jupyter Notebook, Conda, TCP/UDP, Random Forest, LSTM, GRU, CNN, RNN, Neural Networks, Clustering, KNN, Recommender Systems, OpenAPI, Swagger, Express, BGP, OSPF

Tools / Platforms:	Git, VS Code, Docker, Kubernetes, Terraform, Jenkins, Github Actions, REST API, GraphQL, JWT, Okta, Active Directory, Linux, Solid Principles, OOPS, Operating Systems, Distributed Systems, Microservices, Serverless, Kafka, ETL, Spark, Athena, Glue, Puppet, Chef, Ansible, HDFS, PySpark, Pytorch, Nginx, Proxy, Reverse Proxy, Putty, Cisco VPN, Netscaler, Load Balancer, Route53, DNS, Wireshark, RDP, Workspaces, Gitlab, IPv4, IPv6, GitLab
Databases:	SQL, NoSQL, Redis, MongoDB, RDS, DynamoDB, PostgreSQL, Oracle, Cassandra, Vector DB, MySQL, SQL Server, SSIS

Projects / Open-Source

Gartner Publishing System Modernization Project:

- Resolved data synchronization issues and implemented secure third-party integration, enabling consistent timestamps and direct authentication.
- Implemented Dead Letter Queues and SNS notifications, removing 100% manual efforts for fetching errors and providing near real-time updates.

Certifications

- AWS Generative AI Practitioner
- AWS Solutions Architect Professional
- AWS DevOps Engineer Professional
- AWS Security Specialty
- AWS Solutions Architect - Associate
- AWS Developer - Associate
- AWS SysOps Administrator - Associate

Honors & Awards

- Qualified for IIT-JEE MAINS & ADVANCED 2016
- Containers SME at AWS Professional Services
- Member of AWS serverless TFC